

Cheng-Tse Liu (Alex)

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EDUCATION

University of California, Santa Cruz

- Master of Science in Natural Language Processing (Computer Science)
- GPA: 3.9/4.0

Santa Clara, CA
Sep. 2023 - Dec. 2024

National University of Kaohsiung

- Bachelor of Science in Information Management
- GPA: 3.52/4.0, Last 60 credits: 3.8/4.0

Kaohsiung, Taiwan
Sep. 2018 - Jun. 2022

TECHNICAL SKILLS

Python, Java, C, HTML/CSS, PHP, Flutter, Keras, Pytorch, Tensorflow, Scikit-learn, MySQL, FireBase, Deep Neural Networks, Regression/Classification, Machine Learning, NumPy, Pandas, LLMs, BERT, Transformer, NLP, RAG, LangChain, Microsoft Azure

PROFESSIONAL EXPERIENCE

University of California, Santa Cruz

NLP Software Developer

- Designing and implementing new features for the CruzChat platform
- Collaborating with faculty and researchers to explore innovative functionality based on user needs
- Identifying, debugging, and resolving system issues
- Developing testing infrastructure to ensure reliability and maintainability of newly integrated features

Santa Clara, CA
Feb. 2025 - Present

Adobe

Machine Learning Engineer Intern

- Collaborated with researchers to design evaluation protocols and analysis workflows for LLM-based summarization
- Developed algorithms for text segmentation and conducted prompt optimization across diverse model configurations
- Investigated cost-efficiency trade-offs between centralized and decentralized multi-agent strategies
- Contributed to internal research presentations and an conference paper submission

Santa Clara, CA
Mar. 2024 - Dec. 2024

Computational Biology & Intelligence System Lab, National University of Kaohsiung

Research Assistant

- Crawled data from websites and preprocessed data as Natural Language Processing (NLP) model input
- Fine-tuned BERT models to identify the transcription factor-gene pairs association
- Achieved outstanding accuracy of 83%
- Used the ROC curve analysis technique to evaluate the results of different Internal Ribosome Entry Site (IRES) prediction tools

Kaohsiung, Taiwan
Jan. 2021 - Jan. 2022

PROJECT EXPERIENCE

Multi-LLM Text Summarization

- Developed a multi-agent pipeline for efficiently summarizing long documents using state-of-the-art LLMs
- Designed a two-stage chunking pipeline to process documents beyond standard context limits
- Deployed and evaluated LLMs including GPT-4o, Llama3, Claude, Gemini and Mistral
- Achieved up to 2× performance improvement over single-LLM baselines on ROUGE and BLEU metrics

Jan. 2024 - Mar. 2025

JeBERTy

- Fine-tune a pre-trained generative model on answer-questioning Japanese dataset
- Train a BERT model to predicted sentence similarity
- Achieved 0.89 Cosine Similarity for generated questions on test set

Jan. 2024 - Mar. 2024

Semantic Textual Relatedness for African and Asian Languages

- Fine-tuned BERT model on Hausa named entity recognition
- Achieved 0.03 MSE

Sep. 2023 - Jan. 2024

PUBLICATIONS

- Fang, J., **Liu, C. T.**, Kim, J., Bhedaru, Y., Liu, E., Singh, N., ... & Deilamsalehy, H. (2024). Multi-LLM Text Summarization. arXiv preprint arXiv:2412.15487.
- T.-H. Yang*, C.-Y. Wang†, H.-C. Tsai†, Y.-C. Yang† and **C.-T. Liu**, “YTLR: a web service for automatic yeast transcription factor-gene association extraction from the literature” Computational and Structural Biotechnology Journal, vol. 20, pp. 4636-4644, 2022 (SCI 2022 impact factor = 6.155, Ranking 23.6% (70/296) in Biochemistry Molecular Biology)
- T.-H. Yang*, C.-Y. Wang†, H.-C. Tsai†, and **C.-T. Liu†**, “Human IRES Atlas: an integrative platform for studying IRES-driven translational regulation in humans” Database, vol. 2021: article ID baab025; doi:10.1093/database/baab025, 2021. (SCI 2020 impact factor = 3.451, Ranking 24% (14/58) in Mathematical Computational Biology)